AEROLIS REGIONAL AIRCRAFT
Customer Support Engineering · Toulmont Airfield

Category: RECOMMENDED
Models: AR-72-100, AR-72-200

Main Landing Gear – Torque Link Apex Pin Inspection

Service Bulletin SB 32-0147, Revision 1

1. PLANNING INFORMATION

A. Effectivity

This service bulletin applies to AR-72 aircraft, manufacturer serial numbers 0101 through 0388, fitted with torque link assembly P/N D23190-5.

B. Reason

Operators have reported wear of the torque link apex pin bushings beyond the limits of the component maintenance manual. Excessive wear can cause shimmy of the main landing gear during taxi and landing.

C. Compliance

Accomplish the inspection at the next scheduled A-check or within 600 flight hours after receipt of this bulletin, whichever occurs first.

D. Manpower

Task	Persons	Elapsed hours	Man-hours
Access and jacking	2	1.0	2.0
Inspection (both gears)	1	2.5	2.5
Close-up and return to service	2	0.5	1.0
Total		4.0	5.5

2. MATERIAL INFORMATION

Item	New P/N	Description	Qty per A/C	Note
1	D23190-118	Bushing, apex pin	4	Replace if worn
2	NAS1611-012	Packing, preformed	4	Discard old part
3	MS24665-302	Cotter pin	2	–
4	MIL-PRF-81322	Grease, general purpose	A/R	Consumable

3. ACCOMPLISHMENT INSTRUCTIONS

WARNING: MAKE SURE THAT THE GROUND SAFETY LOCKS ARE INSTALLED ON THE LANDING GEAR. Movement of the landing gear can cause injury to persons and damage to the aircraft.

1. Prepare the aircraft.
   a. Put the aircraft on jacks in accordance with AMM 07-11-00.
   b. Remove hydraulic pressure from the brake system.
2. Remove the torque link apex pin.
   a. Remove the cotter pin and the castellated nut.
   b. Support the upper torque link and pull out the apex pin.
3. Measure the bushing inside diameter at two positions 90 degrees apart.
4. Compare the results with the limits in Table 3.
5. Replace the bushings that exceed the limits and record the result in the maintenance computer.

CAUTION: DO NOT USE A HAMMER TO REMOVE THE APEX PIN.

Use the extraction tool listed in the AMM to prevent damage to the bushing bore.

Table 3. Limits and torque values

Location	Inside diameter (mm)		Torque value	
	Nominal	Max. worn	N·m	lbf·in
Upper link bushing	19.050	19.300	34 - 41	300 - 360
Lower link bushing	19.050	19.300	34 - 41	300 - 360
Apex pin nut	–	–	45 - 51	400 - 450

Record format for the maintenance computer

Enter the inspection result in the following format so that the fleet database can be updated automatically.

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TASK      SB32-0147-R1
MSN       0217
GEAR      LH
UPPER_ID  19.112
LOWER_ID  19.087
RESULT    PASS
SIGNED    LIC-B1-40218
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4. RETURN TO SERVICE

Do an operational test of the landing gear retraction system in accordance with AMM 32-30-00. Make an entry in the aircraft technical log that this service bulletin has been accomplished.